# Sam Williams

samhw6@gmail.com | (703) 244-7802 | smwllyms.github.io | github.com/smwllyms

## Education

**M.S. Computer Science, Virginia Tech** January 2023 – December 2023
- Thesis: *Exploring the Usability of Non-verbal Vocal Interaction (NVVI) and a Pitch-based Implementation*

**B.S. Computer Science, Virginia Tech** Fall 2019 – Fall 2022

## Research & Publications

**Distributed Virtual Environments (DVE) Lab, Virginia Tech** Spring 2020 – Spring 2024
*Undergraduate and Graduate Researcher*
- Investigated **non-verbal vocal interaction (NVVI)** as a human-computer interaction modality, designing and evaluating pitch-based implementations as continuous control mechanisms.
- Applied **audio signal-processing** techniques, including pitch analysis, **autocorrelation**, and the **Fourier transform**, to develop interactive audio systems.
- *A Relative Pitch Based Approach to Non-verbal Vocal Interaction as a Continuous and One-Dimensional Controller*, HCII 2024.
- *An Approach to Pitch Based Implementation of Non-verbal Vocal Interaction*, IEEE VR 2024.
- *Immersive Technology in the Public School Classroom: When a Class Meets*, iLRN 2021.

## Featured Projects

**Audio Plugin Sandbox**
- Built a web-based environment for quickly and interactively prototyping audio plugins and signal-processing systems.
- Translated C/C++-based audio processing programs into JavaScript using the Web Audio API.
- Designed a drag-and-drop interface for routing generators and effects, with real-time parameter controls such as sliders.

**VST Production**
- Developed audio plugins in C++ using a VST SDK, including pitch/timbre shifting, filtering, and compression.
- Applied signal-processing concepts and Model-View-Controller (MVC) principles to audio plugin development.

**X86 ASM Sandbox/Debugger**
- Built an interactive x86 assembly debugger with terminal and GUI interfaces, breakpoints, and visualization/editing of the stack, registers, flags, and RAM.

## Work Experience

**Senior Platform Engineer, Oracle** January 2025 – Present
- Developed, maintained, and deployed UI for six high-traffic Oracle Cloud Infrastructure (OCI) plugins using Node.js, TypeScript, React, and Playwright; managed features, bugs, and library updates across service, product, and UX teams, and regularly presented UX features and patterns to service teams, Product Managers, and UX review boards.
- Co-created a new multi-edit UX pattern for bulk data management, now used in the OCI Full Stack Disaster Recovery service.
- Architected and implemented a Node.js-based MCP server that automatically generates tools for RESTful HTTP operations from OpenAPI specifications, supports thousands of tools through discovery-based invocation, and enables custom service API configurations.
- Designed and developed a comprehensive Chrome extension for UI/UX debugging and testing, providing request redirection, data mocking, payload/header modification, and configurable network behavior through `declarativeNetRequest` and runtime script injection. Developed an agentic SDLC framework around these capabilities enabling Codex to implement features, debug, test, and perform regression autonomously using structured prompting and guardrails for functional integrity.

**Instructor of Record, Virginia Tech** January 2024 – May 2024
- Taught an introductory Graphical User Interfaces course to 70 students, covering GUI programming, computer graphics, and full-stack development through technologies including MVC, React, MUI, WebGL, D3, and the MERN stack; developed course materials and delivered lectures in a fast-paced environment.
- Supervised and coordinated teaching assistants, organizing weekly meetings and establishing procedures for office hours and grading.

**Software Engineer, ARIES Program, Virginia Tech Libraries** May 2021 – August 2021
- Developed a cross-platform web application for desktop and VR devices, translating Figma UI designs into JavaScript/Three.js experiences.

## Technical Skills

**Programming Languages:** TypeScript, JavaScript, HTML, C, C++, C#, Python, SQL, Java, Rust, x86 Assembly

**Audio & Music:** Audio/Digital Signal Processing, Music Production, Audio Recording, Sound Design, Audio Plugins, VST, Web Audio API, Pitch Analysis

**Web & Software:** React, Node.js, REST APIs, Testing, Playwright, APIs, UI/UX, MVC, MERN, WordPress

**Technologies:** Unity3D, Three.js, Blender, WebGL, SVG, D3, Virtual/Extended Reality, Agentic AI, IDEs